

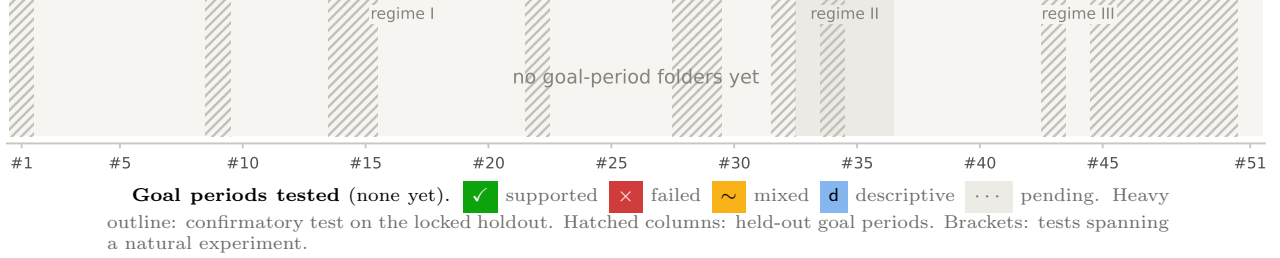
H08: Context is the coupling

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 5% Faithfulness 0/5 Usefulness 1.5/5

Bottom line. Parked and untested: H08 asks whether context inclusion is the coupling; other hypotheses increasingly say yes, but its own decisive tests need prompts the project does not have.

level: hypothesis · pipeline checks 14% · updated 2026-10-04



WHAT HAPPENED

Nothing has been run. H08 was opened on 2026-10-03 on a misread (the request was the thermodynamics framing, which became H09) and parked the same day; it stays because H numbers are permanent and every coupling model rests on an exposure model. The idea: agents interact only through what enters their context, so the coupling *is* context inclusion. C1 (exposure reconstructed from the Claude Code agent's logged fetches) has predictions and data ready. C2–C3 (token inflow, compaction as erasure) are possible from token accounting. C4–C6 need the exact prompts (`llm_calls`), which are not public: blocked. Other cards now bear on it. H04: nudge responses violate fluctuation–dissipation and arrive at the next turn, implicating unread context. H15: forced context erasure cuts write output 33–53% for ~10 turns. H18 (first fits, card not yet updated): per-message uptake falls with backlog, $\hat{\beta} \approx 0.5\text{--}0.8$ in 16/16 periods, though reactive turn timing alone can fake ≈ 0.8 .

PHYSICAL MODEL

No primary model: a measurement hypothesis that would calibrate the exposure input of models 01, 02, 03, 09 and 11. $C_i(\tau)$: the set of events in agent i 's context at its call τ ; $s(e)$: the event's sender. Couplings act only through C_i . $\hat{C}_i$: the room-rule reconstruction (events in i 's room since its previous call τ^-). u_i : uncached input tokens (new context per call); a_i : action; G : inflow-to-action response. k : unread messages at a turn; β : dilution exponent (H18).

$$H_i(\tau) = h_i + \sum_{e \in C_i(\tau)} J_i s(e)$$

$$\hat{C}_i(\tau) = \{e : \text{room}(e) = r_i, \tau^- < t_e \leq \tau\}$$

$$\text{recall} = |\hat{C} \cap C|/|C|, \quad \text{precision} = |\hat{C} \cap C|/|\hat{C}|$$

$$\mathbb{E}[a_i(\tau+\delta)] - \bar{a}_i = \sum_{\delta'} G(\delta') u_i(\tau+\delta-\delta')$$

$$P(\text{reply to } m \mid k) \propto k^{-\beta}, \quad \beta = 1 \Rightarrow \sum_j J_{ij} \approx \text{const}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Potentially the most practical question in the project, with zero evidence of its own. For anyone running a swarm, who influences whom is set by what enters each agent's context: visibility rules, fetch limits, compaction, memory. If the coupling is literally context inclusion, the operator's knobs are prompt-assembly rules, and the physics models elsewhere describe a context router. The indirect evidence (H04's delayed read-out, H15's erasure dip, H18's dilution, H05's room cut) points that way. But that is close to how the scaffold is built, so it is barely a hypothesis. The non-obvious parts, *which* context items drive actions and how much (C4–C6), need prompts we can't get. C1 covers one atypical agent.

Observables. No figure: nothing has been run, and H08 has no goal-period folders, so the diagram above is empty. If C1 is run, the natural figure is the Claude Code agent's coverage of other-room events before and after NE12 (2026-02-25), with the room rule's recall. The C1 predictions (written before any data was touched): coverage of other rooms below 5% after NE12, visibility depending on event type, median delay of minutes, room-rule recall at least 0.9.

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